

Library System Analysis

Practice Questions & Solutions

Q1. Create a New Book Record — insert '978-1-60129-456-2', 'To Kill a Mockingbird', 'Classic', 6.00, 'yes', 'Harper Lee', 'J.B. Lippincott & Co.'

Solution:

```
INSERT INTO
books(isbn,book_title,category,rental_price,status,author,publisher)
VALUES ('978-1-60129-456-2', 'To Kill a Mockingbird', 'Classic',
6.00, 'yes', 'Harper Lee', 'J.B. Lippincott & Co.');
```



```
SELECT * FROM books;
```

Q2. Update an Existing Member's Address

Solution:

```
UPDATE members
SET member_name = 'Alice John', member_address = '321 Main St'
WHERE member_id IN ('C101');
```



```
SELECT * FROM members;
```

Q3. Delete the record with issued_id = 'IS121' from the issued_status table.

Solution:

```
DELETE FROM issued_status
WHERE issued_id = 'IS121';
```



```
SELECT * FROM issued_status;
```

Q4. Retrieve All Books Issued by a Specific Employee — emp_id = 'E101'.

Solution:

```
SELECT
    i.issued_book_name,
    i.issued_date,
    i.issued_book_isbn,
    e.emp_name
```

```
FROM issued_status i JOIN employee e ON e.emp_id = i.issued_emp_id
WHERE i.issued_emp_id IN ('E101');
```

Q5. List Members Who Have Issued More Than One Book

Solution:

```
SELECT
    m.member_id,
    m.member_name,
    COUNT(i.issued_emp_id) AS total_book_issued
FROM issued_status i LEFT JOIN members m ON m.member_id =
i.issued_member_id
GROUP BY 1,2
HAVING COUNT(i.issued_emp_id) > 1
ORDER BY 3 DESC;
```

Q6. Create Summary Tables — use CTAS to generate a new table showing each book and total times issued.

Solution:

```
CREATE TABLE book_count AS
SELECT
    b.book_title,
    COUNT(b.isbn) AS total_book_issued
FROM issued_status i LEFT JOIN books b ON i.issued_book_isbn = b.isbn
GROUP BY 1
ORDER BY 2 DESC;

SELECT * FROM book_count;
```

Q7. Retrieve All Books in a Classic Category

Solution:

```
SELECT
    *
FROM books
WHERE category IN ('Classic');
```

Q8. Find Total Rental Income by Category

Solution:

```
SELECT
    b.category,
    COUNT(i.issued_book_isbn) AS total_book_issued,
    SUM(b.rental_price) AS total_income
FROM issued_status i LEFT JOIN books b ON b.isbn = i.issued_book_isbn
GROUP BY 1;
```

Q9. List Employees with Their Branch Manager's Name and Branch Details

Solution:

```
WITH CTE AS
(
    SELECT
        e.*,
        b.manager_id
    FROM employee e LEFT JOIN branch b ON e.branch_id = b.branch_id
    WHERE e.position NOT IN ('Manager')
)
SELECT
    c.*,
    e.emp_name
FROM CTE c JOIN employee e ON c.manager_id = e.emp_id;
```

Q10. Create a Table of Books with Rental Price Above the Average Threshold

Solution:

```
CREATE TABLE book_price AS
SELECT
    *
FROM books
WHERE rental_price > (SELECT AVG(rental_price) FROM books);

SELECT * FROM book_price;
```

Q11. Retrieve the List of Books Not Yet Returned

Solution:

```
SELECT
    i.issued_id,
    i.issued_book_name,
    i.issued_date,
    i.issued_book_isbn,
    r.return_id,
    r.return_date
FROM return_status r RIGHT JOIN issued_status i ON r.issued_id =
i.issued_id
WHERE r.return_id IS NULL;
```